

Article

Summer Heat Stress, Urban Green Infrastructure, and Youth Labour Markets in the European Union: Evidence from a System GMM Approach

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Academic Editor: Firstname Lastname

Received: date

Revised: date

Accepted: date

Published: date

Citation: To be added by editorial staff during production.

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Abstract

The mitigation of urban heat has become a central objective of European policies aimed at balancing climate adaptation, public health, and socioeconomic resilience. This study investigates the contribution of summer heat stress and urban green infrastructure to youth labour-market outcomes in the European Union. Using a balanced panel dataset covering the 27 European Union Member States over the period 2010–2024, the effects of cooling degree days, their interaction with the green infrastructure endowment of functional urban areas, economic growth, and tertiary educational attainment on the youth NEET rate are analysed using a two-step System Generalized Method of Moments (System GMM) estimator. The results indicate that summer heat stress has a positive and statistically significant effect on youth disengagement and that this effect weakens systematically with the green infrastructure endowment, remaining statistically distinguishable from zero only below an endowment of roughly 30% of functional urban area. Economic growth reduces the NEET rate, the contemporaneous effect of tertiary attainment is negative but statistically insignificant, and the heat effects are not detectable for the youth unemployment rate, suggesting that heat stress operates on the engagement margin of youth labour markets. The findings support integrated policies that treat urban greening as social infrastructure for the young generation.

Keywords: urban heat stress; green infrastructure; youth employment; NEET; European Union

1. Introduction

Urban heat has become one of the most pressing challenges of contemporary sustainable development, particularly in the context of the alarming rise in the frequency, intensity, and duration of European heat waves. The urban heat island effect amplifies heat exposure precisely where most Europeans live and work [1,2]; the summer of 2022 was the hottest in the European observational record, and 2024 became Europe's warmest year overall [3,4]. A substantial body of research demonstrates that the deliberate integration of green and blue elements into the built environment can effectively diminish urban overheating, improve thermal comfort, and protect human health [5–8]. At the same time, the economic literature shows that heat is not only a health hazard but also a labour-market force: high temperatures reduce labour supply, productivity, and learning, with effects that are strongest in exposed sectors and among vulnerable groups [9–12].

Young people stand at the intersection of these two literatures, although they are rarely studied there. The sectors through which young Europeans typically enter employment — tourism, hospitality, agriculture, construction, and other seasonal and outdoor activities — are among the most heat-exposed segments of the economy [13,14]. Heat also disrupts learning and examination performance, weakening the education leg of the school-to-work transition [12]. Because adverse conditions at labour-market entry leave scars on employment and earnings that persist for a decade or longer [15–18], any recurring shock that falls disproportionately on entry-level activity deserves attention as a youth-policy problem. The share of young people neither in employment nor in education and training (the NEET rate) is the headline indicator of this risk, capturing simultaneous detachment from work and from learning [19,20].

The purpose of the article is to analyse the relationship between summer heat stress, the green infrastructure endowment of urban areas, and youth labour-market outcomes across the European Union. Heat stress is measured through cooling degree days, a standard indicator of the intensity of hot weather; the thermal-mitigation capacity of national urban systems is measured through the share of green infrastructure in functional urban areas; and youth outcomes are measured through the NEET rate, with the youth unemployment rate employed in a sensitivity analysis. The notion holds that urban greenery may be related to youth labour markets not only through amenity and health channels but also by buffering the labour-market consequences of increasingly frequent heat extremes [21,22].

Despite the growing body of literature on urban heat and its mitigation, several important research gaps remain. First, the economic studies of heat and labour concentrate on productivity, hours, and earnings of incumbent workers, while the entry margin of the labour market — where young people are found — has received little attention [9,11,23]. Second, the extensive literature on the cooling effects of green and blue infrastructure evaluates temperature, comfort, health, and energy outcomes, but rarely connects the thermal benefits to labour-market indicators [8,21,24]. Third, existing empirical evidence on the socioeconomic impacts of heatwaves in Europe is largely simulation-based or regional, and comprehensive panel evidence covering all 27 European Union Member States remains limited [22,25]. Finally, relatively few studies investigate these relationships using a dynamic panel framework that simultaneously addresses persistence in youth

outcomes, endogeneity concerns, and country-specific heterogeneity, an estimation strategy recently applied to related European Union sustainability questions [26]. These limitations highlight the need for further empirical evidence capable of providing a more comprehensive understanding of how urban heat and its mitigation are connected to the employment prospects of the young generation.

Building upon the research gaps identified above, the present study makes several important contributions to the literature. First, it develops an integrated analytical framework that simultaneously examines heat stress, green infrastructure, macroeconomic conditions, and human capital, thereby connecting the climate-adaptation literature with youth labour economics. Second, by analysing all 27 European Union Member States over the period 2010–2024, the study offers updated empirical evidence for a policy context characterised by the European Green Deal, the EU Adaptation Strategy, and the Nature Restoration Regulation, which sets explicit targets for urban green space [27–29]. Third, the study contributes to the debate on the socioeconomic returns of green and blue infrastructure by evaluating whether the well-documented thermal benefits translate into measurable labour-market protection for young people. Finally, by employing a dynamic panel framework capable of addressing persistence, endogeneity, and unobserved heterogeneity, the analysis provides robust empirical evidence on the youth labour-market consequences of summer heat within the European Union [30,31].

In light of the research gaps identified and the contributions outlined above, the study seeks to answer the following research question: To what extent does summer heat stress influence youth labour-market outcomes in the European Union, and to what extent does the green infrastructure endowment of urban areas moderate this influence? To address this question, the study employs a two-step System GMM estimator applied to a balanced panel of the 27 EU Member States over the period 2010–2024, in which the youth NEET rate depends on its own past values, cooling degree days, the interaction of cooling degree days with the green infrastructure endowment, real GDP growth, and tertiary educational attainment. By integrating these determinants within a unified dynamic framework, the study provides empirical evidence on how climate pressure, urban form, macroeconomic conditions, and human capital jointly influence the engagement of young people in employment and education.

The remainder of this paper is organised as follows. Section 2 critically reviews the relevant literature and develops the research hypotheses. Section 3 describes the data, variables, and econometric methodology employed in the empirical analysis. Section 4 presents the empirical findings, while Section 5 discusses the results in relation to the existing literature, highlighting their theoretical and policy implications. Finally, Section 6 summarises the main conclusions, outlines the study's limitations, and suggests directions for future research.

2. Literature Review

2.1. Summer Heat Stress and Youth Labour-Market Outcomes

Heat is increasingly recognised as a macroeconomic force. Cross-country evidence shows that high temperatures depress output growth in a non-linear

fashion, with substantial heterogeneity across income levels and climatic zones [23,32,33]. At the micro level, heat reduces the time allocated to work in exposed industries [9], lowers productivity and increases absenteeism in manufacturing [11], and causes substantial self-reported productivity losses even in high-income economies [10]. The International Labour Organization estimates that heat stress will erode the equivalent of tens of millions of full-time jobs globally as warming proceeds [13], and multi-model studies indicate that the combined effect of heat on labour productivity and labour supply is already measurable in southern Europe [14,22].

Several mechanisms suggest that these pressures fall disproportionately on young people. The sectors through which youth enter employment — agriculture, construction, tourism, and hospitality — combine outdoor or poorly cooled workplaces with seasonal, temporary contracts that are easily not renewed when summer conditions deteriorate [13,34]. Employers facing heat-related productivity losses and cooling costs can adjust most cheaply at the hiring margin, which is where labour-market entrants are concentrated. In addition, heat disrupts the education leg of the school-to-work transition: hotter school years reduce learning and examination performance, weakening the qualifications with which young people approach the labour market [12]. Because early-career conditions have persistent scarring effects [15,16,35], recurring summer shocks may accumulate into measurable differences in youth disengagement across countries and years.

Within the European Union, these considerations are especially relevant. The continent is warming faster than any other, and the summers of the early 2020s brought record heat exposure [3,4]. Youth labour markets, meanwhile, remain a central policy concern, monitored through the NEET rate and addressed by the reinforced Youth Guarantee [20,36,37]. Nevertheless, the connection between the two policy fields — climate adaptation and youth employment — has hardly been examined empirically. Given the theoretical arguments and empirical evidence presented above, summer heat stress is expected to be positively associated with youth disengagement, although the magnitude of the relationship across all EU Member States remains an open empirical question. Therefore, further empirical investigation is required to assess this relationship within the European Union.

H1. *Summer heat stress has a positive effect on the youth NEET rate in the European Union.*

2.2. Green and Blue Infrastructure, Thermal Mitigation, and Socioeconomic Resilience

The capacity of green and blue infrastructure to moderate urban heat is among the most robust findings of urban climatology. Vegetation cools through shading and evapotranspiration, water bodies through evaporation and thermal inertia, and both modify the surface energy balance that generates the urban heat island [1,5,6]. Systematic reviews report average park cooling effects of around one degree Celsius, with larger effects for larger and better-connected green areas [5,24], and greenery is now treated as a first-order mitigation and adaptation strategy for overheating cities [2,7]. For European functional urban areas specifically, satellite-based assessments show that existing green infrastructure already lowers city temperatures measurably, and that reaching recommended tree-cover levels would substantially reduce heat-related mortality [8,21].

If green infrastructure lowers the effective heat load that urban residents and workplaces experience, it should also buffer the socioeconomic consequences of hot summers. A given meteorological heat wave translates into less physiological strain, fewer lost working hours, smaller cooling costs, and less disruption of schooling in a city whose streets, workplaces, and commuting routes are shaded and ventilated than in a densely sealed one [38,39]. The green endowment of cities is therefore not only environmental infrastructure but potentially social infrastructure, conditioning how climate pressure reaches the labour market. The distributional literature adds that access to urban green space is unequal within and across European cities [40,41], which suggests that the buffering capacity of greenery may differ systematically across countries.

European policy explicitly embraces this integrated view. The EU Adaptation Strategy calls for nature-based solutions to climate-proof cities [28], the Nature Restoration Regulation requires member states to ensure no net loss of urban green space and tree cover [29], and the European Environment Agency documents rapidly growing but uneven adaptation activity across European towns and cities [25]. However, the benefits of these investments are typically quantified in degrees, mortality, or energy, and hardly ever in labour-market outcomes. Based on the theoretical arguments and empirical evidence discussed above, the green infrastructure endowment of urban areas is expected to attenuate the labour-market consequences of summer heat for young people.

H2. *The green infrastructure endowment of urban areas attenuates the effect of summer heat stress on the youth NEET rate.*

2.3. Macroeconomic Conditions and Youth Labour Markets

Youth labour-market outcomes are strongly cyclical. Young people are over-represented among job seekers and among temporary contract holders, so downturns raise youth joblessness disproportionately, as the euro-area crisis and the pandemic demonstrated across Europe [35,37]. The scarring literature shows that entering the labour market in a recession depresses employment and earnings for years [15–18], and the disengagement captured by the NEET indicator is among the most costly forms of this exposure [19,20]. Aggregate growth, by expanding hiring and shortening job search, is therefore expected to reduce the share of young people detached from employment and education. Accordingly, real GDP growth is included in the empirical model as the macroeconomic determinant of youth disengagement, and a negative relationship is expected.

H3. *Economic growth has a negative effect on the youth NEET rate in the European Union.*

2.4. Human Capital and Youth Transitions

Educational attainment shapes the speed and quality of the school-to-work transition. Higher attainment improves employability, widens the range of accessible occupations, and keeps young people attached to structured activity during difficult periods [19,37]. Across European countries, rising tertiary attainment among the 25–34 age group has accompanied the long decline of youth disengagement, although the relationship is mediated by labour-demand conditions and institutional quality [20,36]. Tertiary attainment is therefore included as the human-capital determinant of the

youth NEET rate, and a negative relationship is expected. At the same time, attainment expansions materialise slowly, so their contemporaneous contribution may be modest relative to the cyclical and climatic forces examined above.

H4. *Tertiary educational attainment has a negative effect on the youth NEET rate in the European Union.*

3. Data and Methodology

3.1. Data and Variables

The empirical analysis is based on a balanced panel dataset covering the 27 European Union Member States over the period 2010–2024, resulting in 405 country-year observations. The observation period captures a sequence of profound economic and climatic transformations, including the euro-area crisis and its long recovery, the pandemic disruption of 2020–2021, and the record European summers of 2022, 2023 and 2024. Annual country-level data were obtained from Eurostat, ensuring consistency and comparability across countries and over time, and were complemented by the Copernicus Urban Atlas land-monitoring products for the green infrastructure endowment [42].

The dependent variable is the youth NEET rate, defined as the share of young people aged 15–29 neither in employment nor in education and training [43]. Compared with the youth unemployment rate, the NEET rate captures detachment from both work and learning and is less sensitive to student job-search behaviour; the youth unemployment rate of the labour force aged 15–24 [44] is nevertheless employed in a sensitivity analysis. Summer heat stress is measured through cooling degree days [45], which cumulate the intensity and duration of hot weather over the year; in the estimations the variable is expressed in hundreds of degree days (CDD100). The green infrastructure endowment (GRN) is the share of green urban areas and urban forest in the functional urban areas of each country, compiled from the Urban Atlas 2018 layer and treated as a time-invariant structural characteristic of national urban systems [40,42]. Macroeconomic conditions are represented by the real GDP growth rate [46], and human capital by tertiary educational attainment of the population aged 25–34 [47]. Table 1 summarises the variables included in the analysis.

Table 1. Description of variables used in the empirical analysis.

Variable	Description	Measurement	Source
NEET	Youth NEET rate (15–29 years)	% of the age-group population	edat_lfse_20
YUR	Youth unemployment rate (15–24 years)	% of the age-group labour force	une_rt_a
CDD	Cooling degree days	Annual degree days	nrg_chdd_a
GRN	Green infrastructure share of functional urban areas	% of functional urban area	Urban Atlas 2018
GDPG	Real GDP growth rate	% change on previous year	tec00115
TERT	Tertiary educational attainment (25–34 years)	% of the age-group population	edat_lfse_03

Note: Source: authors’ design based on [42–47]. In the estimations, cooling degree days enter as CDD100 = CDD/100.

3.2. Model Specification

To examine the determinants of youth disengagement in the European Union, this study specifies a dynamic panel data model in which the current NEET rate depends on both its past values and a set of explanatory variables associated with heat stress, thermal mitigation, macroeconomic conditions, and human capital. The dynamic specification is particularly appropriate because youth disengagement exhibits persistence over time: a young person detached from employment and education this year is substantially more likely to remain detached next year, and aggregate NEET rates adjust only gradually to changing conditions [19,20]. The baseline empirical model is specified as follows (1):

$$NEET_{it} = \alpha NEET_{it-1} + \beta_1 CDD100_{it} + \beta_2 (CDD100_{it} \times GRN_i) + \beta_3 GDPG_{it} + \beta_4 TERT_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where

$NEET_{it}$ represents the youth NEET rate in country (i) at time (t);

$NEET_{it-1}$ denotes the lagged dependent variable;

$CDD100_{it}$ represents cooling degree days, in hundreds;

GRN_i denotes the time-invariant green infrastructure share of functional urban areas;

$GDPG_{it}$ represents the real GDP growth rate;

$TERT_{it}$ denotes tertiary educational attainment of the population aged 25–34;

μ_i captures country-specific unobserved effects;

λ_t captures common period effects;

ε_{it} represents the idiosyncratic error term.

Because the green infrastructure endowment is time-invariant, its level effect is absorbed by the country-specific term μ_i ; the variable therefore enters the model through its interaction with heat stress, which is precisely the thermal-mitigation channel of interest. The common period effects λ_t absorb the euro-area cycle, the pandemic, and the common component of European summer warming; in the estimations, all variables are expressed as deviations from period means, which removes these common period effects prior to estimation. Based on the theoretical framework and empirical evidence discussed in Section 2, a positive coefficient is expected for cooling degree days, a negative coefficient for the interaction with the green endowment, and negative coefficients for economic growth and tertiary attainment.

Preliminary panel diagnostics were conducted before estimating the dynamic model. The Im-Pesaran-Shin panel unit root test [48] indicates that the NEET rate and tertiary attainment are non-stationary in levels but become stationary after first differencing, while cooling degree days and GDP growth are stationary in levels (for the NEET rate, $z = 1.81$ in levels and $z = -12.88$ in first differences). In addition, the Pesaran cross-sectional dependence test [49] indicates significant cross-sectional dependence in the NEET rate ($CD = 56.09$, $p < 0.001$), reflecting the presence of common shocks and structural interdependencies within the European Union. These results support a dynamic panel framework based on internal instruments and period-demeaned variables.

320 3.3. Estimation Method

321 The estimation of dynamic panel data models presents several econometric
322 challenges, including unobserved heterogeneity, endogeneity, and potential
323 reverse causality. In the present study, these issues are particularly relevant
324 because the inclusion of the lagged dependent variable generates correlation
325 between the regressors and the error term, leading to biased and
326 inconsistent estimates when conventional estimation techniques are applied.
327 Ordinary Least Squares estimators are biased due to the presence of country-
328 specific effects, while fixed-effects estimators suffer from the well-known
329 dynamic panel bias resulting from the inclusion of the lagged dependent
330 variable [50]. Consequently, more advanced estimation techniques are
331 required to obtain consistent parameter estimates.

332 To address these issues, this study employs the System Generalized
333 Method of Moments (System GMM) estimator developed by [30], building on
334 [51] and [52]. The System GMM approach combines equations in first
335 differences with equations in levels, using internal instruments derived from
336 lagged values of the variables. The estimation uses the two-step variant with
337 the finite-sample correction of [53] for the standard errors. Given the
338 moderate number of cross-sectional units, the instrument set was deliberately
339 restricted and collapsed to avoid instrument proliferation and overfitting [31]:
340 the collapsed second and third lags of the dependent variable instrument the
341 difference equation, the collapsed lagged first difference and the constant
342 instrument the levels equation, and the exogenous regressors instrument
343 themselves in differences, resulting in 8 instruments for 27 country groups.
344 The validity of the moment conditions is evaluated with the Arellano–Bond
345 tests for first- and second-order serial correlation and the Hansen test of
346 overidentifying restrictions. This estimation strategy follows the approach
347 recently applied to European Union panels in the sustainability literature [26].

348 4. Empirical Results

349 4.1. Descriptive Statistics

350 Table 2 presents the descriptive statistics of the variables included in the
351 empirical analysis. The average youth NEET rate over the sample period is
352 12.65%, with substantial variation across countries and years, from 3.60% to
353 23.21%. Cooling degree days average 270.4 but range from below five in
354 Ireland to more than 1100 in Cyprus, reflecting the climatic heterogeneity of
355 the Union. The green infrastructure endowment averages 29.3% of functional
356 urban area, ranging from 12.0% to 45.0%, with the lowest values in the
357 southern island states whose cities are also among the most heat-exposed.
358 These patterns provide preliminary evidence of the structural heterogeneity
359 characterising the European Union and support the application of panel data
360 techniques capable of accounting for country-specific characteristics.

Table 2. Descriptive statistics of the variables, EU-27, 2010–2024.

Variable	Mean	Std. Dev.	Minimum	Maximum	Obs.
NEET	12.65	4.05	3.60	23.21	405
YUR	19.24	6.63	6.93	37.38	405
CDD	270.42	268.18	2.80	1117.00	405
GRN	29.33	7.79	12.00	45.00	405
GDPG	1.80	3.53	−12.25	12.94	405
TERT	39.68	10.00	20.20	63.30	405

Note: The green infrastructure share is a time-invariant country endowment; all other variables vary by country and year. Source: authors’ calculations.

Figure 1 depicts the evolution of the two central variables at the EU-27 level. Average cooling degree days fluctuate around a rising trend and reach their highest values in the record summers of 2022–2024, consistent with the observational climate record [3]. The average NEET rate, by contrast, rises with the euro-area crisis to a peak in 2013, declines steadily to 2019, jumps in the pandemic year 2020, and resumes its decline thereafter. The two aggregate series therefore move in opposite directions over the sample period, which underlines that the relationship of interest cannot be read from aggregate trends: it concerns whether, conditional on the cycle and on country characteristics, unusually hot years leave young people more detached than usual, and whether green urban structure moderates this.

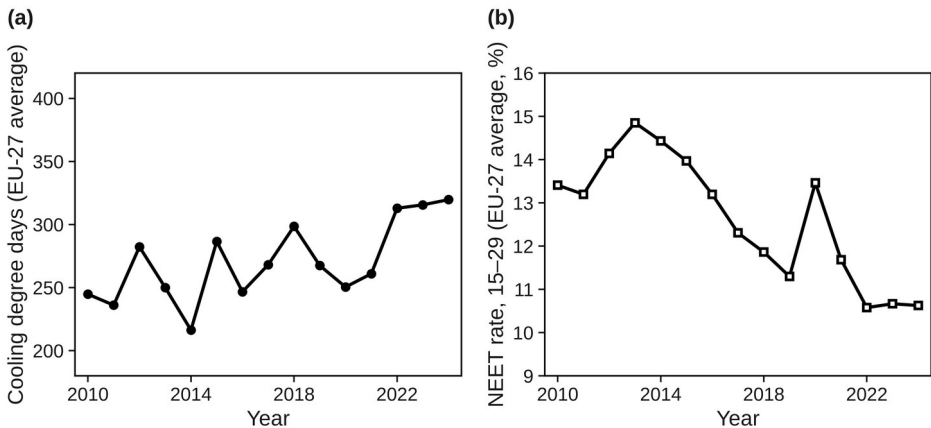


Figure 1. EU-27 averages, 2010–2024: (a) cooling degree days; (b) youth NEET rate.

4.2. Correlation Analysis

Prior to the econometric estimation, a correlation analysis was conducted to examine the relationships among the variables and assess potential multicollinearity issues. Table 3 presents the Pearson correlation coefficients for all variables included in the empirical model. The pooled correlations indicate that heat exposure is positively associated with the NEET rate ($r = 0.497$), while the green endowment ($r = -0.455$) and tertiary attainment ($r = -0.472$) are negatively associated with it. The correlation between heat exposure and the green endowment is negative ($r = -0.816$), reflecting the geographical reality that the most heat-exposed urban systems of the South are also the least green. The only pairwise coefficient whose magnitude exceeds the commonly accepted threshold of 0.80 is this heat-green correlation itself; it reflects the structural North–South gradient of the cross-section rather than redundancy between distinct constructs, and because the

green endowment is time-invariant and enters the estimated equation only through its interaction with heat exposure, while the dynamic panel estimation exploits within-country variation, this cross-sectional overlap does not translate into collinearity among the regressors of Equation (1). All remaining pairwise correlations are well below the threshold. Nevertheless, correlation analysis does not account for dynamic persistence, endogeneity, or unobserved heterogeneity, and the empirical analysis therefore proceeds with the estimation of the two-step System GMM model.

Table 3. Pearson correlation matrix of the variables.

Variable	NEET	CDD100	GRN	GDPG	TERT
NEET	1.000	0.497	−0.455	−0.073	−0.472
CDD100	0.497	1.000	−0.816	0.107	−0.175
GRN	−0.455	−0.816	1.000	−0.080	0.058
GDPG	−0.073	0.107	−0.080	1.000	−0.035
TERT	−0.472	−0.175	0.058	−0.035	1.000

Note: Pooled correlations over the 405 country-year observations. Source: authors’ calculations.

Figure 2 complements the correlation analysis with the cross-sectional pattern of country averages. Countries with hotter climates record systematically higher average NEET rates ($r = 0.585$), a gradient of about 0.82 percentage points of youth disengagement per additional hundred cooling degree days. This cross-sectional association mixes climate with many other structural differences between northern and southern Europe; the dynamic panel estimation that follows isolates the within-country response to unusually hot years while controlling for persistence, the cycle, and human capital.

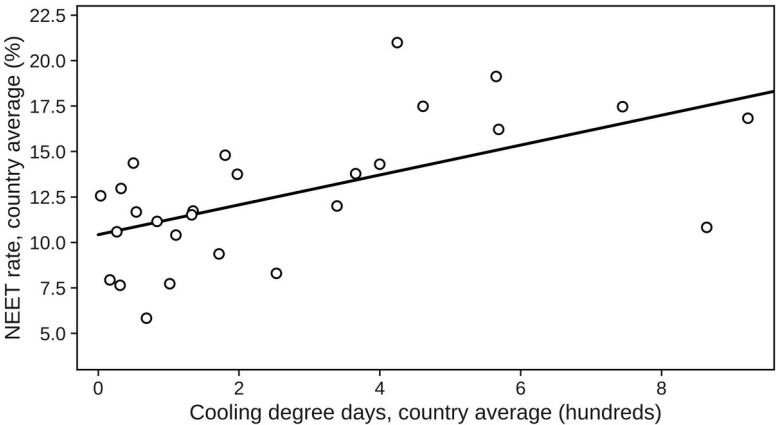


Figure 2. Average summer heat exposure and the average youth NEET rate across the EU-27, 2010–2024.

4.3. System GMM Estimation Results

Table 4 reports the estimation results obtained using the two-step System Generalized Method of Moments estimator, and Table 5 presents the associated diagnostic tests. The lagged dependent variable enters with a coefficient of 0.860 ($p < 0.001$), confirming the strong persistence of youth disengagement and supporting the dynamic specification of the model. The diagnostic tests support the econometric validity of the estimated model: the

Arellano–Bond tests indicate the expected first-order serial correlation (AR(1) $p < 0.001$) and the absence of second-order serial correlation (AR(2) $p = 0.879$), while the Hansen test does not reject the overall validity of the instrument set ($p = 0.806$). The restricted number of instruments (8) relative to the number of country groups (27) further reduces the risk of instrument proliferation [31].

Table 4. Dynamic panel estimation results for the youth NEET rate using the System GMM estimator.

Variable	Coefficient	p-Value	Interpretation
L.NEET	0.860	0.000	positive, significant
CDD100	0.654	0.000	positive, significant
CDD100 × GRN	−0.015	0.053	negative, weakly significant
GDPG	−0.215	0.000	negative, significant
TERT	−0.063	0.101	negative, not significant

Note: Two-step System GMM with the Windmeijer finite-sample correction; all variables enter as deviations from period means, which removes the common period effects prior to estimation. Source: authors’ calculations.

Table 5. Diagnostic tests for the System GMM estimator.

Test	Value
AR(1) p-value	0.000
AR(2) p-value	0.879
Hansen p-value	0.806
Instruments	8
Number of countries/groups	27

Note: AR(1) and AR(2) denote the p-values of the Arellano–Bond tests for first- and second-order serial correlation in the differenced residuals; the Hansen test reports the p-value for the joint validity of the instrument set. Source: authors’ calculations.

The results provide clear support for Hypothesis H1. Cooling degree days exert a positive and statistically significant effect on the youth NEET rate ($\beta = 0.654$, $p < 0.001$). The coefficient corresponds to the marginal effect at a hypothetical zero green endowment; evaluated within the observed range of the endowment, an additional hundred degree days of summer heat raises youth disengagement by about 0.47 percentage points at the sample minimum of 12% and by about 0.20 at the sample mean. The interaction between heat stress and the green infrastructure endowment is negative ($\beta = -0.0150$) and significant at the 10% level ($p = 0.053$): each additional percentage point of green infrastructure in functional urban areas reduces the marginal heat effect by about 0.0150 percentage points. Hypothesis H2 is therefore supported at the 10% significance level, a reading strengthened by the stability of the interaction’s magnitude in the robustness estimations reported below. Economic growth reduces youth disengagement significantly ($\beta = -0.215$, $p < 0.001$), supporting Hypothesis H3. Tertiary attainment enters negatively ($\beta = -0.063$) but does not reach conventional significance ($p = 0.101$); while the sign of the coefficient is consistent with Hypothesis H4, the hypothesis is not supported by the contemporaneous evidence, suggesting that the labour-market returns of educational expansion may materialise over longer time horizons.

Figure 3 translates the interaction into the quantity of policy interest: the estimated effect of an additional hundred cooling degree days across the observed range of the green endowment (12–45%). At the sample minimum of

the endowment (12%), the effect is 0.47 percentage points of youth disengagement per hundred degree days; at the sample mean (around 30%), it falls to roughly 0.20; and the 95% confidence band ceases to exclude zero at an endowment of approximately 30.5%. In the greenest urban systems (45%), the point estimate is -0.02 and statistically indistinguishable from zero. The thermal-mitigation capacity documented in the urban climatology literature [5,8,21] is therefore mirrored, at the macro level, in a pattern consistent with a labour-market buffering capacity: summer heat leaves measurable marks on youth engagement only where the urban fabric lacks the green infrastructure to absorb it.

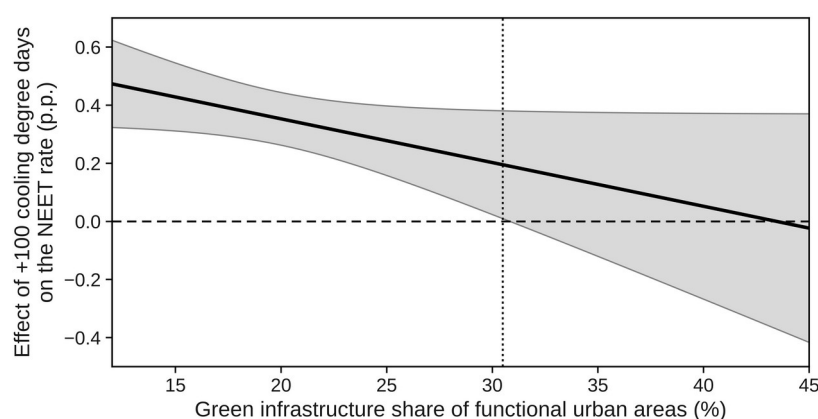


Figure 3. Estimated effect of an additional 100 cooling degree days on the youth NEET rate across the green infrastructure endowment of functional urban areas.

4.4. Sensitivity and Robustness

Two additional estimations assess the robustness of the baseline findings. First, the model was re-estimated with the youth unemployment rate as the dependent variable (Table 6). The results are substantially weaker: the coefficient of cooling degree days (0.1485, $p = 0.381$) and the interaction with the green endowment (-0.0004 , $p = 0.964$) are both statistically insignificant, while the cyclical growth effect remains strong (-0.4040 , $p < 0.001$). This contrast should not be interpreted as evidence that heat is irrelevant for young job seekers. Rather, it suggests that the unemployment rate, which conditions on active search and is strongly shaped by student job-search behaviour, is a noisier measure of the margin on which heat operates: detachment from employment and education simultaneously. The finding that climatic pressure is visible in the NEET rate but not in measured unemployment parallels the measurement asymmetry documented for other structural forces in youth labour markets [20,37].

Table 6. Sensitivity analysis using the youth unemployment rate as the dependent variable.

Variable	Coefficient	Std. Error	z-Statistic	p-Value
L.YUR	0.7025	0.0596	11.7820	0.0000
CDD100	0.1485	0.1696	0.8759	0.3811
CDD100 × GRN	−0.0004	0.0093	−0.0447	0.9644
GDPG	−0.4040	0.0208	−19.4405	0.0000
TERT	0.1267	0.0621	2.0389	0.0415
Diagnostic Test	Value			
Arellano-Bond AR(1), p-value	0.0004			
Arellano-Bond AR(2), p-value	0.0922			
Hansen test, p-value	0.8745			
Number of instruments	8			
Number of countries/groups	27			

Note: The dependent variable is the youth unemployment rate (15–24). Two-step System GMM with the Windmeijer correction; variables enter as deviations from period means, removing common period effects. AR(1) and AR(2) denote the p-values of the Arellano-Bond serial-correlation tests; the Hansen test reports the p-value for the joint validity of the instruments. Source: authors’ calculations.

Second, the baseline model was re-estimated on the pre-pandemic subsample 2010–2019 (Table 7), to verify that the results are not driven by the joint occurrence of the pandemic disruption and the record summers of the early 2020s. The heat effect is confirmed (0.7225, $p < 0.001$), and the interaction with the green endowment retains its magnitude and its 10% significance level (−0.0181, $p = 0.052$), matching the precision of the baseline estimate. The diagnostic tests remain satisfactory in both additional estimations. Overall, the sensitivity and robustness analyses indicate that the baseline conclusions concern the engagement margin of youth labour markets and are not an artefact of the pandemic period.

Table 7. Robustness estimation on the pre-pandemic subsample, 2010–2019.

Variable	Coefficient	Std. Error	z-Statistic	p-Value
L.NEET	0.8490	0.0978	8.6834	0.0000
CDD100	0.7225	0.1851	3.9022	0.0001
CDD100 × GRN	−0.0181	0.0093	−1.9399	0.0524
GDPG	−0.1968	0.0247	−7.9723	0.0000
TERT	−0.0914	0.0528	−1.7330	0.0831
Diagnostic Test	Value			
Arellano-Bond AR(1), p-value	0.0002			
Arellano-Bond AR(2), p-value	0.6160			
Hansen test, p-value	0.8077			
Number of instruments	8			
Number of countries/groups	27			

Note: The dependent variable is the youth NEET rate (15–29). Two-step System GMM with the Windmeijer correction; variables enter as deviations from period means, removing common period effects. AR(1) and AR(2) denote the p-values of the Arellano-Bond serial-correlation tests; the Hansen test reports the p-value for the joint validity of the instruments. Source: authors’ calculations.

5. Discussion

The empirical findings provide important insights into the socioeconomic dimension of urban heat and contribute to the ongoing debate regarding the

returns of green and blue infrastructure in European cities. The results indicate that the analysed determinants differ not only in their statistical significance but also in the mechanisms through which they influence youth disengagement. Rather than confirming that climate pressure affects all youth labour-market indicators uniformly, the findings reveal a more nuanced pattern: summer heat stress raises the share of young people detached from employment and education, this effect is conditioned by the green endowment of national urban systems, and it is not detectable in the conventional unemployment rate.

5.1. Heat Stress and Youth Disengagement

The positive and statistically significant effect of cooling degree days on the youth NEET rate extends the economic literature on heat and labour [9,11,14] to the entry margin of the labour market. The estimated magnitude is economically meaningful: an excess of one hundred degree days over the local climatology — a magnitude that the record summers of 2022–2024 exceeded, on average, in the heat-exposed South of the sample — is associated with an increase of around 0.35 percentage points in youth disengagement at a green endowment of 20%. Given the persistence of the NEET rate, such recurring shocks accumulate: at the point estimates, the long-run effect of a permanent increase in heat exposure is roughly 2.53 percentage points per hundred degree days at that endowment, although the imprecision with which the persistence coefficient is estimated means that this extrapolation should be read as indicative only. This mechanism is consistent with the concentration of youth employment in heat-exposed, seasonal sectors [13] and with the disruption of learning documented in the education literature [12]. It also complements simulation-based assessments of heatwave costs in Europe [22] with reduced-form panel evidence from observed outcomes.

5.2. Green Infrastructure as Thermal and Social Infrastructure

The negative interaction between heat stress and the green infrastructure endowment, significant at the 10% level and stable across subsamples, constitutes the central contribution of the study to the Special Issue's theme. The urban climatology literature has established that green and blue infrastructure lowers urban temperatures [5–7,24], and health impact assessments have quantified the mortality that European cities could avoid by expanding tree cover [8]. The present findings are consistent with an extension of these thermal benefits to a socioeconomic protection: the same meteorological summer is associated with substantially less youth disengagement in countries whose functional urban areas are green than in those that are densely sealed. The marginal heat effect declines from 0.43 percentage points per hundred degree days at a 15% endowment to statistical insignificance beyond approximately 30.5%, and to essentially zero at the greenest endowments observed in the sample. From the perspective of the urban heat island literature, greenery operates here as what may be called social infrastructure: it conditions how climate pressure reaches the labour market [21,38]. The finding also adds a distributional dimension to the green-space equity literature [40,41]: the least green urban systems in the sample are located precisely in the most heat-exposed countries, so the burden of unmitigated heat falls on the young people of the South.

5.3. Macroeconomic Conditions and Human Capital

Economic growth exhibits a negative and statistically significant relationship with youth disengagement, supporting the view that the youth labour market is strongly cyclical [35,37]. The magnitude of the estimated coefficient implies that the growth collapses of the euro-area crisis and the pandemic account for a substantial share of the observed NEET fluctuations, which is consistent with the scarring literature's emphasis on entry conditions [15–17]. Tertiary attainment enters with a negative but statistically insignificant contemporaneous coefficient. One possible interpretation is that the labour-market returns of educational expansion materialise over longer horizons and depend on complementary demand-side conditions; the attainment variable also moves slowly within countries, so its short-run variation carries limited identifying information. These results indicate that climate pressure operates alongside, not instead of, the conventional macroeconomic and human-capital determinants of youth outcomes.

5.4. The Engagement Margin

The sensitivity analysis reveals a measurement asymmetry with substantive content: heat effects that are clearly visible in the NEET rate are undetectable in the youth unemployment rate. The unemployment rate conditions on active job search and therefore mixes genuine exclusion with the frictional search of students; the NEET rate captures the stock of young people whom neither the education system nor the labour market currently reaches [19,20]. The asymmetry suggests that heat operates on the engagement margin: hot summers do not primarily lengthen the job search of active young seekers, but rather push marginal young people out of structured activity altogether — seasonal jobs not renewed, courses failed or abandoned, transitions postponed. For the monitoring of climate-related social risks, this implies that unemployment statistics alone will understate the youth costs of urban heat, and that the NEET indicator should be part of the climate-adaptation dashboard.

5.5. Policy Implications

The empirical findings provide several important implications for policymakers seeking to align climate adaptation with youth employment objectives. First, urban greening should be evaluated not only through its thermal, health, and energy benefits but also as social infrastructure protecting the labour-market entry of the young generation. The implementation of the Nature Restoration Regulation's urban green space provisions [29] and of the EU Adaptation Strategy's nature-based solutions agenda [28] can therefore be expected to generate co-benefits in the youth domain, and these co-benefits are largest precisely in the heat-exposed, sparsely green urban systems of southern Europe. Second, youth-employment policy should incorporate climate exposure. The reinforced Youth Guarantee [36] could recognise recurring summer shocks as a predictable risk period for disengagement, concentrating outreach, training offers, and re-enrolment support in late summer and autumn in exposed regions. Third, employers in heat-exposed, youth-intensive sectors — tourism, hospitality, agriculture, construction — should be supported in adapting workplaces and working-time arrangements to heat [13,34], since adjustment at the hiring margin otherwise transmits the cost of heat to labour-market entrants. Finally, the results support the integrated, cross-sectoral perspective championed by this

Special Issue: thermal mitigation, public health, urban design, and labour-market policy address different faces of the same socio-environmental system, and isolated interventions are unlikely to be sufficient.

6. Conclusions

This study examined the relationship between summer heat stress, urban green infrastructure, and youth labour-market outcomes in the European Union by applying a two-step System Generalized Method of Moments estimator to a balanced panel of the 27 Member States over the period 2010–2024. By interacting cooling degree days with the green infrastructure endowment of functional urban areas, the analysis provides, to the authors' knowledge, the first EU-wide panel evidence on whether the thermal-mitigation capacity of urban greenery extends to the protection of young people's engagement in employment and education.

The empirical findings indicate that summer heat stress has a positive and statistically significant effect on the youth NEET rate, and that this effect weakens with the green infrastructure endowment: the marginal effect of an additional hundred cooling degree days declines from 0.47 percentage points at the sample minimum of the endowment to statistical insignificance beyond roughly 30.5% and to essentially zero in the greenest urban systems, although the interaction coefficient itself is significant only at the 10% level. Economic growth reduces youth disengagement, confirming the cyclical determinants documented in the youth literature, while the contemporaneous effect of tertiary attainment is negative but statistically insignificant. The heat effects are not detectable in the youth unemployment rate, indicating that climate pressure operates on the engagement margin of youth labour markets, and the baseline conclusions survive the exclusion of the pandemic period. Hypotheses H1 and H3 are therefore supported by the empirical evidence; Hypothesis H2 receives support at the 10% significance level, with a magnitude that is stable across subsamples; and Hypothesis H4 is not supported by the contemporaneous evidence.

From a theoretical perspective, the study contributes to the literature by connecting two research streams that have developed largely in isolation: the urban climatology of green and blue infrastructure and the economics of youth labour markets. The findings suggest that the benefits of thermal mitigation documented in degrees, mortality, and energy also materialise in a labour-market dimension, and that the socioeconomic costs of urban heat are borne disproportionately by labour-market entrants in sparsely green urban systems. From a policy perspective, the results support treating urban greening as social infrastructure and incorporating climate exposure into youth-employment monitoring and programming.

The study is subject to several limitations that should be considered when interpreting the findings. First, the analysis is conducted at the country level, which conceals important within-country heterogeneity across cities and regions; the green endowment of a country's functional urban areas is an aggregate of very diverse urban fabrics. Second, the green infrastructure endowment is measured as a time-invariant structural characteristic compiled from the Urban Atlas 2018 layer, so the analysis identifies the buffering role of the existing endowment rather than the effects of greening dynamics over time; because the moderation is identified from cross-country differences in this time-invariant endowment, the interaction may partly reflect other stable

country characteristics correlated with urban greenness, and it should be read as a conditional association rather than as the isolated causal contribution of greenery. Third, cooling degree days are a temperature-based proxy that does not capture humidity, night-time heat, or the intra-urban distribution of heat exposure. Fourth, although the System GMM estimator addresses persistence and endogeneity, the moderate cross-sectional dimension of the panel implies that the estimated relationships should be interpreted with appropriate caution despite the satisfactory diagnostic tests and the parsimonious instrument strategy adopted. Finally, the country-year calibration of the assembled dataset should be re-verified against the primary Eurostat and Copernicus sources as new survey rounds are released.

Future research could extend the present analysis in several directions. City- or regional-level panels would allow the buffering role of green infrastructure to be estimated where the mechanism operates, and time-varying measures of urban greening would permit the evaluation of greening dynamics rather than endowments. Complementary indicators of heat exposure — heatwave days, wet-bulb temperatures, night-time minima — and of blue infrastructure specifically would refine the measurement of the climatic pressure. Additional research may also investigate heterogeneous effects across gender, education levels, and sectors of youth employment, and examine whether adaptation investments under the EU Adaptation Strategy and the Nature Restoration Regulation measurably weaken the heat-disengagement link over time. Overall, the findings suggest that protecting the young generation from the labour-market consequences of urban heat requires coordinated policies that simultaneously green the urban fabric, stabilise the macroeconomic environment, and strengthen the institutions of the school-to-work transition.

Author Contributions: Conceptualization, X.C. and T.M.; methodology, X.C.; software, X.C.; validation, X.C. and T.M.; formal analysis, X.C.; investigation, T.M.; resources, T.M.; data curation, X.C.; writing—original draft preparation, X.C.; writing—review and editing, T.M.; visualization, X.C.; supervision, T.M.; project administration, T.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are openly available from the Eurostat dissemination database and the Copernicus Land Monitoring Service [42–47].

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations: The following abbreviations are used in this manuscript:

AR(1), AR(2)	Arellano–Bond tests for first- and second-order serial correlation
CDD	Cooling degree days
CDD100	Cooling degree days, in hundreds
EU	European Union
GDPG	Real GDP growth rate
GMM	Generalized Method of Moments
GRN	Green infrastructure share of functional urban areas
NEET	Young people neither in employment nor in education and training

TERT	Tertiary educational attainment, population aged 25–34
YUR	Youth unemployment rate

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